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A Functionally Self-Testable RISC-V Processor with a Custom Test Instruction Set

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ABSTRACT

Modern processor based systems are typically validated only during manufacturing, leaving them vulnerable to faults that develop during field operation. This work presents a **functionally self-testable RISC-V processor** that enables **on demand hardware validation** through a custom test instruction. **ATPG-generated test patterns**, stored in a reserved memory region, are executed using the native processor datapath under the control of a dedicated Test FSM. A comparator with pass/fail counters evaluates the test responses, while a **PC save-and-restore mechanism** ensures non-intrusive test execution and seamless resumption of normal processor operation. Experimental evaluation on a **single-cycle RISC-V processor** achieved **98.61% stuck-at fault coverage**, successfully detecting **6489 of 6580 faults** with **minimal hardware overhead**, demonstrating an efficient and lightweight solution for improving processor reliability.

Keywords: RISC-V, Self-Test Processor, ATPG, Fault Detection

INTRODUCTION

Problem Statement

- Processor reliability is critical for long-running and mission-critical applications.
- Runtime faults arise from **aging, environmental effects, and manufacturing defects**.
- Conventional hardware BIST incurs additional hardware overhead, while SBST often depends on handcrafted test programs.
- A lightweight, user-invokable self-test mechanism is required for **on-demand processor validation**.

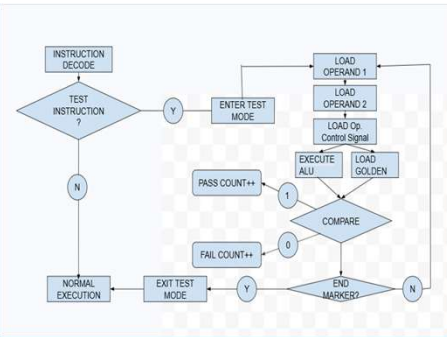
Proposed Solution

- A **functionally self-testable RISC-V processor** is proposed.
- ATPG-generated test patterns are stored in reserved memory and executed through the native datapath.
- A custom test instruction enables **on-demand self-test execution**.
- A dedicated Test FSM manages self-test while preserving normal processor operation.

STATE-OF-THE-ART COMPARISON

Capability	In-Field Processor Validation	On-Demand Test Invocation	Native Instruction Execution	ATPG-Based Test Execution	Automatic Context Preservation	Seamless Execution Recovery
Manufacturing Test	X	X	X	✓	X	X
Hardware BIST	✓	Limited	X	✓	X	Limited
SBST	✓	✓	✓	Limited	Limited	Limited
This Work	✓	✓	✓	✓	✓	✓

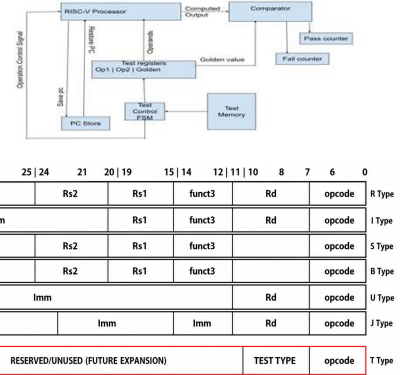
TEST EXECUTION FLOW



The self-test process is initiated when a custom test instruction is decoded. The processor enters test mode, applies ATPG-generated test patterns through the ALU, compares the computed output with the expected golden response, and updates pass/fail counters accordingly. After all test patterns are executed, normal program execution resumes.

PROPOSED ARCHITECTURE

The proposed architecture integrates a custom self-test framework within a single-cycle RISC-V processor. ATPG-generated test patterns are fetched from reserved test memory and coordinated by a dedicated Test Control FSM. Computed outputs are compared against golden responses, while pass and fail counters record test outcomes. A PC save-and-restore mechanism enables non-intrusive testing and seamless return to normal execution.



EXPERIMENTAL SETUP MEMORY ORGANISATION

Processor : Single-Cycle RISC-V
ATPG Tool : TestMax
Fault Model : Stuck-at
Test Patterns : 888
ISA : RV32I
Synthesis Tool : Design Compiler
Test Memory Start : 0x800

0x000	Normal Instruction
0x004	Normal Instruction
0x008	Test Instruction
0x00C	Normal Instruction
...	...
0x800	OPERAND 1
0x804	OPERAND 2
0x808	OPERATION CODE
0x80C	GOLDEN VALUE
...	...
0x3F80	0xFFFFFFFF (END MARKER)

Key Memory Parameters

- ✓ Reserved Test Memory Start : 0x800
- ✓ ATPG Test Patterns : 888
- ✓ End Marker : 0xFFFFFFFF

RESULTS & FAULT COVERAGE

Parameter	Value
Total Faults	6580
Detected Faults	6489
Undetected Faults	91
ATPG Patterns	888
Fault Coverage	98.61%

98.61%

STUCK-AT FAULT
COVERAGE

6489 / 6580

FAULTS DETECTED

The proposed self-test framework achieved 98.61% stuck-at fault coverage, detecting 6489 out of 6580 injected faults using 888 ATPG-generated test patterns.

KEY CONTRIBUTIONS

- ✓ User-triggered runtime self-test
- ✓ Custom RISC-V test instruction support
- ✓ ATPG-driven functional validation
- ✓ Dedicated Test Control FSM
- ✓ PC save-and-restore mechanism
- ✓ Comparator-based result verification
- ✓ Automatic pass/fail status reporting
- ✓ Scalable framework for future processor modules

CONCLUSION

A lightweight functionally self-testable RISC-V processor was developed using a custom test instruction and ATPG-generated test patterns. The proposed framework enables on-demand processor validation through the native datapath while preserving normal execution. Experimental evaluation achieved 98.61% stuck-at fault coverage with minimal hardware overhead.

FUTURE WORK

- Extension to pipelined RISC-V architectures
- Integration with scan-based testing techniques
- FPGA implementation and hardware validation
- Support for delay, transition, and transient fault models